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Identification of *BoRR* gene family in cauliflower: roles in curd development and salt tolerance

Mengfei Song^{1†}, Yusen Shen^{1†}, Jiansheng Wang¹, Huifang Yu¹, Xiaoguang Sheng¹, Shuting Qiao¹ and Honghui Gu^{1*}

Abstract

Background Cauliflower, as an important vegetable crop, the research on its curd formation mechanism and stress-responsive gene networks is of great significance for improving its quality, yield and abiotic stress tolerance. The response regulator (*RR*) gene family plays a crucial role in the regulation of various life processes of many organisms. In this research, a comprehensive analysis of the *BoRR* gene family in cauliflower was carried out.

Results A Total of 57 *BoRR* genes were identified in cauliflower and classified into seven subtypes (type A/B-I/B-II/B-IV/C/B-PRR/Clock PRR) based on sequence homology. Chromosomal mapping showed even distribution across genomes, while physicochemical analysis revealed diverse protein properties (134–915 amino acids, pI 4.51–9.19) with predominant nuclear localization. Structural analyses found all *BoRR* proteins contain REC-type domains, with subtype-specific features: type A has REC_typeA_ARR, type B harbors REC_typeB_ARR domains, and Clock PRR shows circadian-related psREC_RR domains. Exon numbers range from 2 to 10, with type A *BoRR* genes having shorter CDS lengths. Collinearity analysis identified 28 pairs of gene duplicates (26 inter-chromosomal). Comparative analysis showed 133 collinear pairs with *Brassica napus*, 96 with *Brassica rapa*, and only 1 with monocots specie (rice and maize). Promoter analysis identified hormone-responsive motifs (ABRE, TGACG), development-related elements (ARE), and stress-responsive sequences (e.g., MBS for drought tolerance) in the promoters of *BoRR* genes. GO enrichment linked *BoRR* genes to phosphorelay signaling, cytokinin/ethylene response, and developmental processes like meristem maintenance. Expression profiling during curd development showed type A genes (*BoRR23/27/34/38/45*) up-regulated in vegetative-reproductive transition, *BoRR3/6/12/32/54* in curd enlargement, and several genes like *BoRR49* in flower bud differentiation. Salt stress (1.5% NaCl) induced transient expression in 8 of 9 selected *BoRR* genes at day 1 after treatment. qRT-PCR validated their roles in developmental regulation and salt tolerance.

Conclusion This study provides valuable insights into the *BoRR* gene family in cauliflower, laying a foundation for further understanding its genetic mechanisms and potentially guiding efforts to enhance curd quality and salt tolerance in cauliflower.

Keywords Cauliflower, Response regulator, Curd formation, Salt tolerance, Phylogenetic analysis

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Background

Cauliflower (*Brassica oleracea* var. *botrytis*) is a valuable vegetable crop recognized for its nutritional and economic benefits, primarily due to its curd structure. Understanding curd formation mechanism is crucial for improving quality and yield of cauliflower. Recent progress in studying the gene families of cauliflower is providing deeper insights into its genetic architecture and molecular mechanisms. Key gene families include the MADS-box family, which is crucial for floral development and curd formation [1]. The bHLH family, playing an important role in the accumulation of anthocyanins in cauliflower [2]. The PEPB family, maintaining the indeterminate growth of shoot apical meristem, regulating curd development and flowering initiation [3]. These studies on cauliflower curd formation mechanism and gene families have enhanced our understanding of its genetic and molecular aspects. Cauliflower is highly susceptible to multiple abiotic stresses including temperature fluctuations, saline-alkali stress, drought, and heavy metal contamination. These abiotic stress compromise cauliflower growth and productivity by disrupting photosynthesis, damaging cellular membrane integrity, inducing oxidative stress, and disturbing ion homeostasis. Notably, several gene families are involved in regulating stress responses in cauliflower, such as the AP2/ERF transcription factor family, which rapidly responds to salt and drought stress, and its enhanced expression significantly improves plant stress tolerance [4].

The *RR* (*Response Regulator*) gene family plays a crucial role in the regulation of various life processes in a wide range of organisms. Members of the *RR* gene family are widely distributed from bacteria to higher plants. They are mainly involved in signal transduction pathways and are essential for the response to environmental stimuli and the fine-tuning of intracellular physiological processes [5, 6]. In plants, *RR* gene family members are often closely associated with plant hormone signal transduction, photoperiod regulation, and adaptive responses to abiotic stresses such as drought, heat, cold, and salt stress [7–12]. The proteins encoded by these genes usually contain specific domains. Through interaction with other signal molecules, for example, forming a two-component system with sensor kinases, they receive and transmit signals, thereby regulating the expression of downstream genes [13–15]. This affects the growth and development process, morphogenesis, and tolerance ability of plants to various external challenges.

In different plants, such as *Arabidopsis*, rice, and melon, the investigation of *RR* gene families has the potential to unravel the molecular mechanisms underlying key agronomic traits like stress tolerance, flowering time regulation, and yield determination [12, 16–19]. Its complex and orderly regulatory network provides a

solid molecular basis for maintaining the normal growth, development, and survival of organisms. Moreover, The *RR* gene family is likely to influence flowering. *RR* genes are central to cytokinin signal transduction, and changes in their expression can indirectly modify flowering time as seen in *Arabidopsis* [20, 21]. Heterologous expression of the *Arabidopsis* *PRR5* gene in rice can lead to late flowering, increased biomass, and enhanced yield [22]. The C-type *RR* gene *ZaRR24* in *Zanthoxylum armatum*, when ectopically expressed in *Arabidopsis*, suppresses growth, triggers early flowering, and reduces fruit length. Additionally, *ZaRR24* overexpression impacts the expression of A- and B-type *RRs* genes associated with floral meristem and organ characteristics [8].

This paper aims to comprehensively characterize the *BoRR* genes in cauliflower, exploring their gene family members, domain characteristics, and expression features. The findings will provide insights into the molecular mechanisms underlying cauliflower growth, curd development, and salt stress adaptation, facilitating the breeding of varieties with enhanced agronomic performance, quality, and sustainable productivity in saline regions.

Results

Identification, classification, subcellular localization and structure analysis of cauliflower *BoRR* gene family

In the investigation of the *BoRR* gene family within cauliflower, 57 *BoRR* genes were identified, designated as *BoRR1-BoRR57* according to their chromosomal position. These genes are evenly dispersed across various cauliflower chromosomes. Based on homology with *Arabidopsis* proteins, these *BoRR* genes are classified into seven types, namely type A (20), type B-I (1), type B-II (16), type B-IV (4), type C (5), type B-PRR (1) and type Clock PRR (10) (Fig. 1; Table 1). Type A genes, such as *BoRR3*, *BoRR12*, *BoRR13*, and others, are characterized by their relatively shorter amino acid sequences and specific conserved domains. Type B-II genes, including *BoRR2*, *BoRR6*, *BoRR7*, and others, are potentially involved in more intricate signaling cascades. *BoRR5*, *BoRR11*, *BoRR18*, and *BoRR56* which belongs to type B-IV, shows unique sequence motifs that could be associated with specialized functions, perhaps related to cell membrane interactions or specific developmental processes. Type C genes like *BoRR31* and *BoRR32* have their own set of conserved regions that might confer particular regulatory properties. The Clock PRR genes, for instance, *BoRR4*, *BoRR10*, *BoRR20*, etc., are likely to be involved in the regulation of circadian rhythms or time-dependent biological processes, given their similarity to known clock-related genes in other organisms.

Molecular characterization revealed substantial variation among cauliflower *BoRR* proteins: amino acid

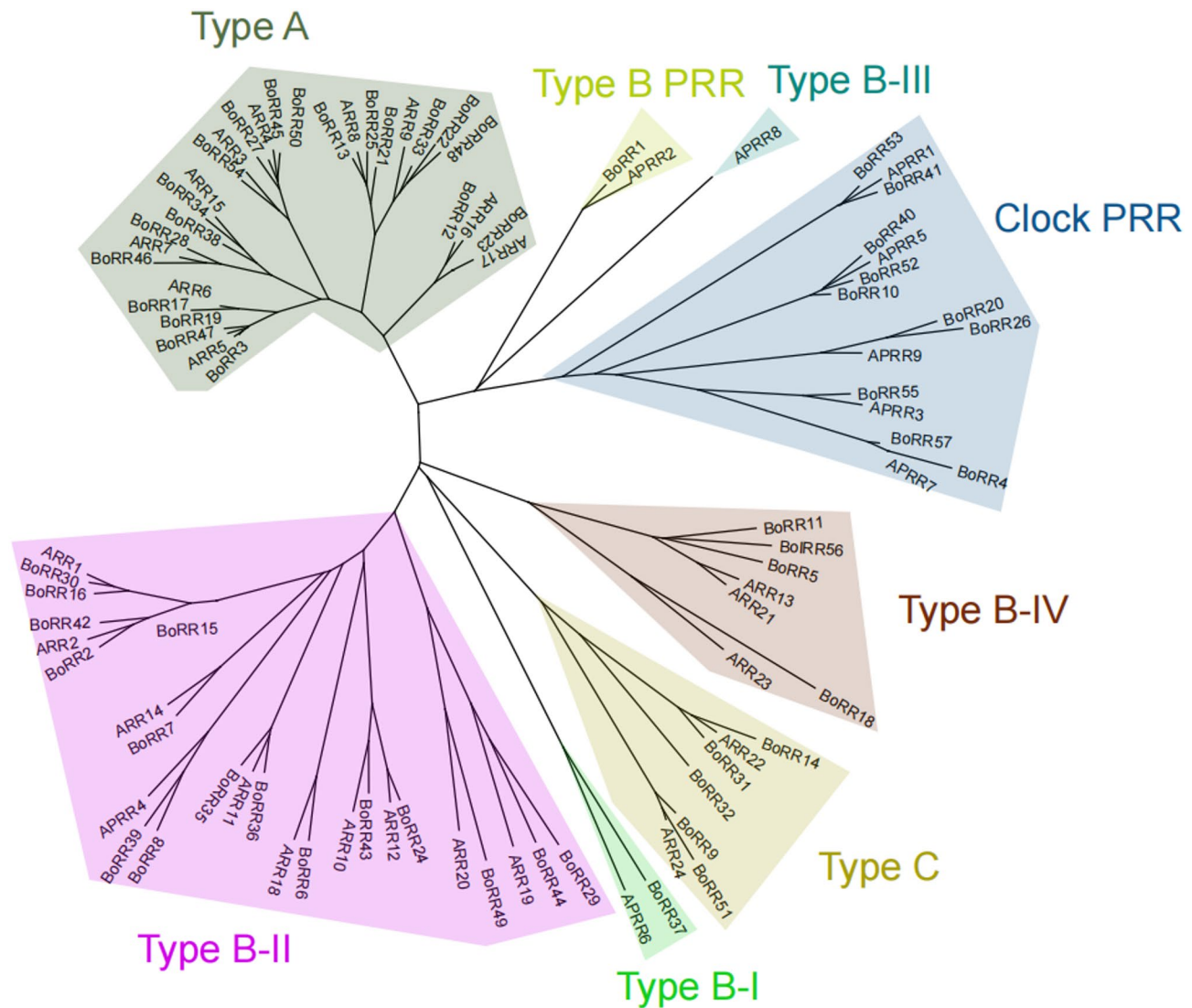


Fig. 1 Evolutionary analysis of the *BoRR* gene family in cauliflower and *Arabidopsis thaliana*. Different colors represent different types of *BoRR* genes: type A, type B-I, type B-II, type B-III, type B-IV, type B PRR, type C and type Clock PRR

lengths ranged from 134 (*BoRR9*, *BolK_2g58340*) to 915 residues (*BoRR8*, *BolK_2g58340*), corresponding To molecular masses of 15.03 kDa and 104.21 kDa, respectively (Table 1; Additional file: Table S1). Isoelectric points exhibited a broad distribution from 4.51 to 9.19 across the family (Table 1). Regarding subcellular localization, a majority of the *BoRR* genes are concentrated in the nucleus (Table 1). Proteins such as *BoRR1*, *BoRR2*, and *BoRR3* are among those with nuclear localization. However, some proteins possess unique localizations. *BoRR5* (*BolK_2g02950*) is situated on the plasma membrane. *BoRR14* (*BolK_3g40150*) is located in the chloroplast. *BoRR8* (*BolK_2g56830*) are found in the endomembrane system.

The structural analysis of the *BoRR* gene family in cauliflower was conducted, including motif analysis, domain

analysis, and gene structure analysis (Fig. 2). The motif analysis of the *BoRR* gene family in cauliflower revealed most of the *BoRR* gene family members possess three types of motifs, namely motif1, motif2, and motif3 (Fig. 2A, Additional file: Table S2). However, significant differences were also observed among different classifications of the *BoRR* gene family. For example, in addition to the commonly existing motifs, the typeB *BoRR* gene family has unique motif9 or motif10. The results of domain analysis showed that all *BoRR* proteins in cauliflower contain the REC-type domain. Further subdivision of different types of *BoRR* gene families revealed their unique domain characteristics (Fig. 2B). The typeA *BoRR* gene family has the REC_typeA_ARR domain, while the typeB has the REC_typeB_ARR or REC_hyHK_CK11_RcsC-like domain, and the Clock PRR *BoRR* proteins have the

Table 1 Information of the *BoRR* gene family in cauliflower

Gene Name	Gene ID	Chr	Position	Group	AA size	Molecular size (kDa)	pI	Subcellular localization
BoRR1	BolK_1g12250	C01	7,176,714–7,180,700	Type B PRR	505	56350.64	6.15	nucleus
BoRR2	BolK_1g26780	C01	18,289,452–18,294,395	Type B-II	631	68971.17	6.08	nucleus
BoRR3	BolK_1g31700	C01	23,816,973–23,818,921	Type A	183	20761.14	5.55	nucleus
BoRR4	BolK_2g00970	C02	667,877–671,518	Clock PRR	539	58678.14	8	nucleus
BoRR5	BolK_2g02950	C02	1,811,038–1,815,799	Type B-IV	826	91196.26	6.23	plasma membrane
BoRR6	BolK_2g15710	C02	9,933,454–9,936,280	Type B-II	626	69897.14	5.37	nucleus
BoRR7	BolK_2g50210	C02	51,359,706–51,361,942	Type B-II	437	48822.87	8.87	nucleus
BoRR8	BolK_2g56830	C02	57,662,745–57,701,930	Type B-II	915	104205.01	7.7	endomembrane system
BoRR9	BolK_2g58340	C02	59,180,067–59,180,666	Type C	134	15030.16	4.82	nucleus
BoRR10	BolK_2g59520	C02	60,066,532–60,068,079	Clock PRR	357	39832.69	6.08	nucleus
BoRR11	BolK_3g03830	C03	1,939,146–1,943,633	Type B-IV	762	85569.1	5.8	nucleus
BoRR12	BolK_3g26350	C03	14,482,039–14,484,027	Type A	158	17693.6	7.11	nucleus
BoRR13	BolK_3g26780	C03	14,782,459–14,783,939	Type A	216	24396.83	4.99	nucleus
BoRR14	BolK_3g40150	C03	24,300,848–24,302,039	Type C	166	18566.8	9.19	chloroplast
BoRR15	BolK_3g48380	C03	29,588,891–29,589,960	Type B-II	157	17549.26	5.7	nucleus
BoRR16	BolK_3g48420	C03	29,610,188–29,613,972	Type B-II	646	70925.29	6.48	endomembrane system
BoRR17	BolK_3g62450	C03	42,374,093–42,375,856	Type A	190	21550.2	6.18	nucleus
BoRR18	BolK_3g63520	C03	43,401,158–43,405,102	Type B-IV	682	74658.61	8.5	nucleus
BoRR19	BolK_3g65310	C03	45,758,649–45,760,211	Type A	192	21927.43	5.7	nucleus
BoRR20	BolK_4g01380	C04	1,651,241–1,653,237	Clock PRR	307	34162.92	7.58	nucleus
BoRR21	BolK_4g03040	C04	2,962,032–2,963,242	Type A	214	24209.91	5.32	nucleus
BoRR22	BolK_4g35010	C04	38,548,899–38,550,429	Type A	232	25762.51	5.03	nucleus
BoRR23	BolK_4g35760	C04	39,156,113–39,157,090	Type A	141	15689.37	6.12	nucleus
BoRR24	BolK_4g54290	C04	56,362,072–56,365,159	Type B-II	581	63913.53	5.98	nucleus
BoRR25	BolK_4g67600	C04	66,189,396–66,190,972	Type A	222	25029.72	5.53	nucleus
BoRR26	BolK_4g71400	C04	68,139,010–68,141,746	Clock PRR	353	38757.8	7.64	nucleus
BoRR27	BolK_5g07030	C05	3,397,121–3,398,550	Type A	253	27483.17	4.65	nucleus
BoRR28	BolK_5g16030	C05	9,150,757–9,152,183	Type A	212	23291.75	5.52	nucleus
BoRR29	BolK_5g32390	C05	27,484,715–27,486,749	Type B-II	550	62293.73	5.57	endomembrane system
BoRR30	BolK_5g48160	C05	47,624,122–47,628,082	Type B-II	680	74075.81	5.94	nucleus
BoRR31	BolK_5g60820	C05	55,924,232–55,925,167	Type C	136	15056.51	5.09	nucleus
BoRR32	BolK_5g60830	C05	55,927,050–55,928,484	Type C	146	16304.66	4.51	nucleus
BoRR33	BolK_6g23480	C06	29,600,315–29,602,114	Type A	184	20520.71	4.86	nucleus
BoRR34	BolK_6g32650	C06	36,550,653–36,552,195	Type A	274	30861.26	5.73	chloroplast
BoRR35	BolK_6g37720	C06	40,884,831–40,887,861	Type B-II	515	58110.88	5.41	nucleus
BoRR36	BolK_6g40340	C06	42,921,358–42,923,431	Type B-II	500	56828.54	5.08	nucleus
BoRR37	BolK_6g41120	C06	43,402,673–43,405,278	Type B-I	521	58252.74	6.38	nucleus
BoRR38	BolK_6g47900	C06	47,819,593–47,821,066	Type A	185	20614.16	5.94	plasma membrane
BoRR39	BolK_7g40200	C07	45,474,884–45,483,005	Type B-II	492	56813.32	5.48	nucleus
BoRR40	BolK_7g43430	C07	47,516,636–47,520,928	Clock PRR	522	57677.35	6.3	nucleus
BoRR41	BolK_7g45370	C07	48,646,299–48,649,257	Clock PRR	573	64350.2	6.73	nucleus
BoRR42	BolK_7g48210	C07	50,288,601–50,291,892	Type B-II	600	65861.78	6.21	nucleus
BoRR43	BolK_7g59040	C07	56,700,434–56,703,139	Type B-II	523	58729.21	5.87	nucleus
BoRR44	BolK_8g03270	C08	2,721,125–2,722,860	Type B-II	450	51257.95	5.41	nucleus
BoRR45	BolK_8g22730	C08	28,671,993–28,673,323	Type A	249	27042.96	4.67	nucleus
BoRR46	BolK_8g27790	C08	31,861,717–31,865,035	Type A	429	48159.1	4.69	nucleus
BoRR47	BolK_8g31770	C08	34,677,046–34,679,118	Type A	183	20745.08	5.83	nucleus
BoRR48	BolK_8g41090	C08	41,458,978–41,460,685	Type A	235	26274.42	5.73	nucleus
BoRR49	BolK_8g46120	C08	44,173,740–44,175,394	Type B-II	419	48035.87	5.12	nucleus
BoRR50	BolK_8g57390	C08	50,093,254–50,094,773	Type A	266	28876.71	4.58	nucleus
BoRR51	BolK_9g05740	C09	3,608,638–3,611,585	Type C	155	17627.37	5.84	cytoplasm
BoRR52	BolK_9g06520	C09	4,093,323–4,096,487	Clock PRR	504	56458.28	8.65	nucleus
BoRR53	BolK_9g07550	C09	4,702,834–4,705,930	Clock PRR	523	59289.59	8.09	nucleus

Table 1 (continued)

Gene Name	Gene ID	Chr	Position	Group	AA size	Molecular size (kDa)	pI	Subcellular localization
BoRR54	BolK_9g21240	C09	16,562,518–16,563,944	Type A	238	26097.08	4.84	nucleus
BoRR55	BolK_9g50460	C09	51,350,968–51,354,192	Clock PRR	471	51542.81	5.71	nucleus
BoRR56	BolK_9g65730	C09	61,420,060–61,424,304	Type B-IV	772	85939.83	5.65	nucleus
BoRR57	BolK_9g70320	C09	63,803,607–63,807,392	Clock PRR	701	76350.09	6.82	nucleus

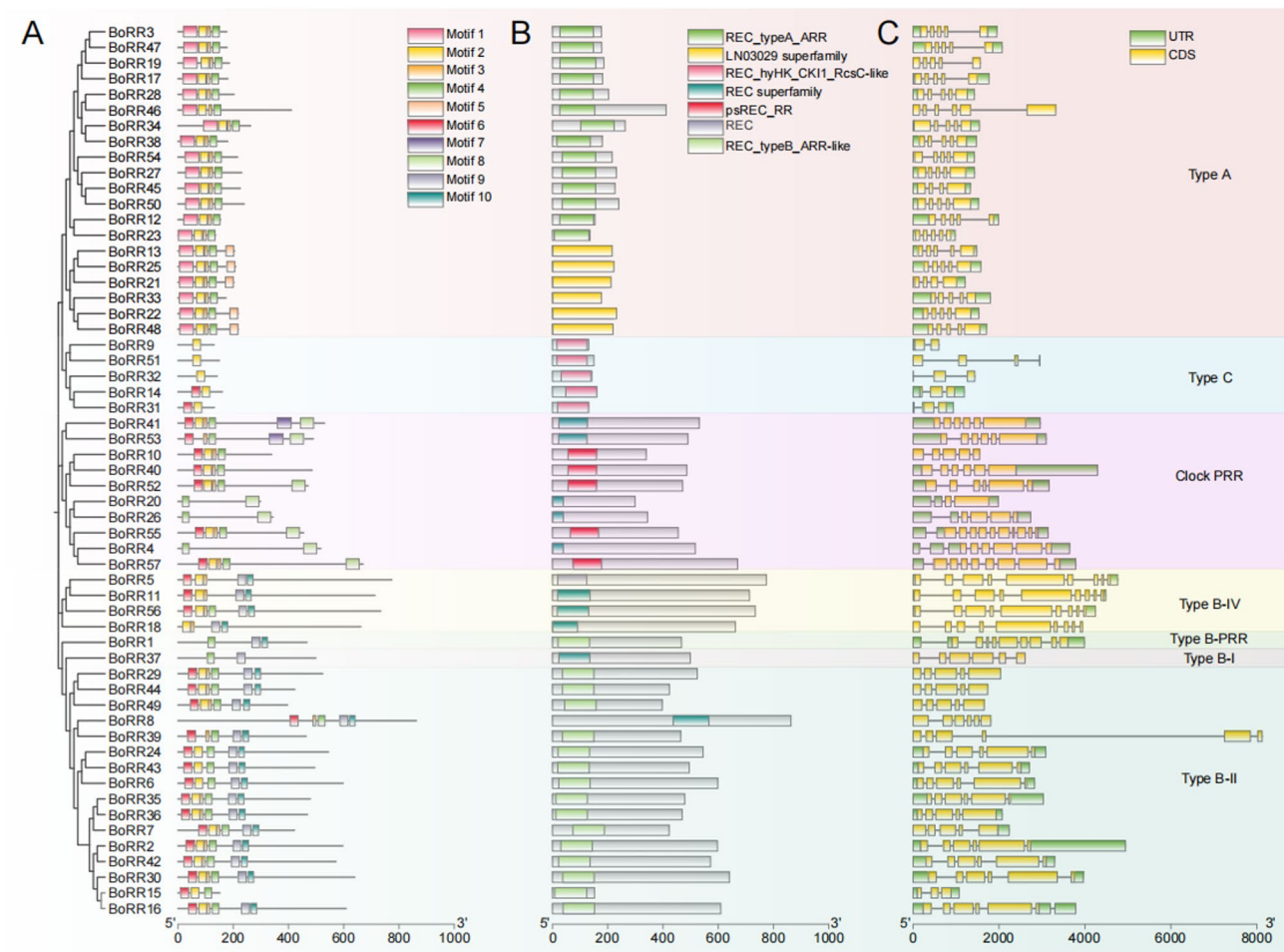


Fig. 2 Analysis of motifs, conserved domains and gene structures of the *BoRR* gene family. **(A)** Ten motifs predicted in the *BoRR* genes by the MEME software are each assigned a distinct color. **(B)** Conserved domains analysis of *BoRR* genes. Different domain features are indicated by different colors. **(C)** Gene structures of each *BoRR* genes. The UTR is represented by blue boxes, the CDS sequence is represented by yellow boxes, and the intron regions are represented by black lines

REC_superfamily or psREC_RR domain. The gene structure analysis indicated that the genes of the *BoRR* gene family in cauliflower exhibit a certain diversity in the number of exons, with the exon number ranging from 2 to 10 (Fig. 2C). It is worth noting that although there is no significant difference in the exon number among different types of *BoRR* gene families, the CDS (coding sequence) and total gene length of the typeA *BoRR* are relatively shorter.

Collinearity analysis of cauliflower *BoRR* gene family

Collinearity analysis of cauliflower *BoRR* genes was carried out (Fig. 3). It was found that a Total of 48 cauliflower *BoRR* genes exhibited collinearity, with 28 pairs of collinear relationships. Notably, 26 pairs of the collinear genes were across chromosomes (Fig. 3A). In contrast, only two pairs of genes showed collinearity within chromosomes. Some genes, such as *BolK3g03830* which is collinear with *BolK2g02950* and *BolK9g65730*, have one-to-many collinear relationships. This one-to-many collinearity is probably due to the triploidization of the *Brassica* genus genome.

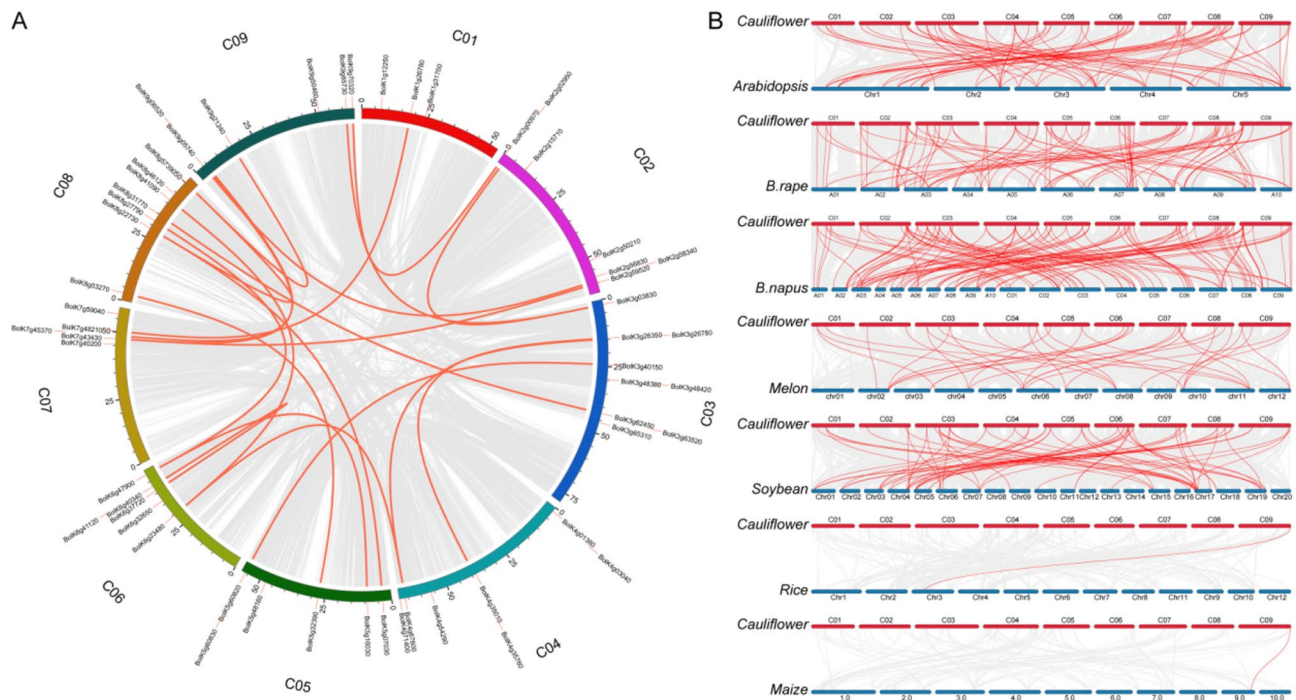


Fig. 3 Collinearity analysis of the *BoRR* gene family in cauliflower. **(A)** Collinearity analysis of the *BoRR* gene family in the cauliflower genome. Different colors of the circle graph represent different chromosomes of cauliflower. The *BoRR* genes are labeled outside the circle. The gray lines indicate the collinearity analysis of all cauliflower genes, and the red lines indicate the collinearity analysis of the *BoRR* genes. **(B)** Collinearity analysis of the *BoRR* gene family in cauliflower with those in *Arabidopsis thaliana*, *B. rapa*, *B. napus*, melon, soybean, rice and maize. The gray lines indicate the collinearity analysis of all genes, and the red lines indicate the collinearity analysis of the *BoRR* genes

The collinearity of the *BoRR* gene family in cauliflower with that of other species was analyzed (Fig. 3B). Among the cruciferous plants, *Arabidopsis thaliana*, *Brassica rapa*, and *Brassica napus* were included. Additionally, we also considered plants from different families such as *Glycine max*, *Cucumis melo*, as well as *Oryza sativa* and *Zea mays* which belong to the monocotyledon group. The results of the collinearity analysis revealed that cauliflower *BoRR* genes share the highest number of collinear genes with *Brassica napus*, precisely 133 pairs. With *Brassica rapa*, the number of collinear gene pairs is 96. When it comes to *Arabidopsis thaliana*, the number of collinear genes is 75 pairs. For *Glycine max*, there are 71 pairs of collinear genes. In comparison with *Cucumis melo*, which has a more distant evolutionary connection to cauliflower, the Total number of evolutionary gene pairs is 40. Interestingly, when contrasted with the monocotyledonous plants *Oryza sativa* and *Zea mays*, only a single pair of genes exhibits collinearity. This indicates that the *BoRR* genes in cauliflower have evolved in a manner that is more closely related to those in other dicotyledonous plants within the same or related genera, while showing a significant divergence from the *BoRR* genes in monocotyledonous plants.

Functional characterization of BARR genes via promoter element and GO enrichment analyses in cauliflower

Through the analysis of the promoter elements of the *BoRR* gene family in cauliflower, a diverse distribution pattern and its potential functional significance have been revealed (Fig. 4). In terms of hormone responsiveness, the promoters of Clock PRR type *BoRR* genes are rich in elements related to hormone response such as ABRE, GGTC, TGACG. In the development-related category, most of the *BoRR* genes in cauliflower carry ARE elements, and the promoters of Clock PRR type *BoRR* genes contain a relatively large variety of development-related elements. In terms of environmental stress, type B-PRR and other types of genes possess drought-responsive elements such as MBS. In the aspect of light responsiveness, some type A genes have multiple light-responsive elements, which may have an important impact on light-related physiological processes such as photomorphogenesis in cauliflower.

Meanwhile, we utilized GO enrichment analysis to interpret the potential functions of cauliflower *BoRR* gene family. The analysis revealed that the *BoRR* gene family is likely involved in several signal transduction pathways, such as the phosphorelay signal transduction pathway and circadian rhythm pathways, along with hormonal response signal pathways including those of cytokinin

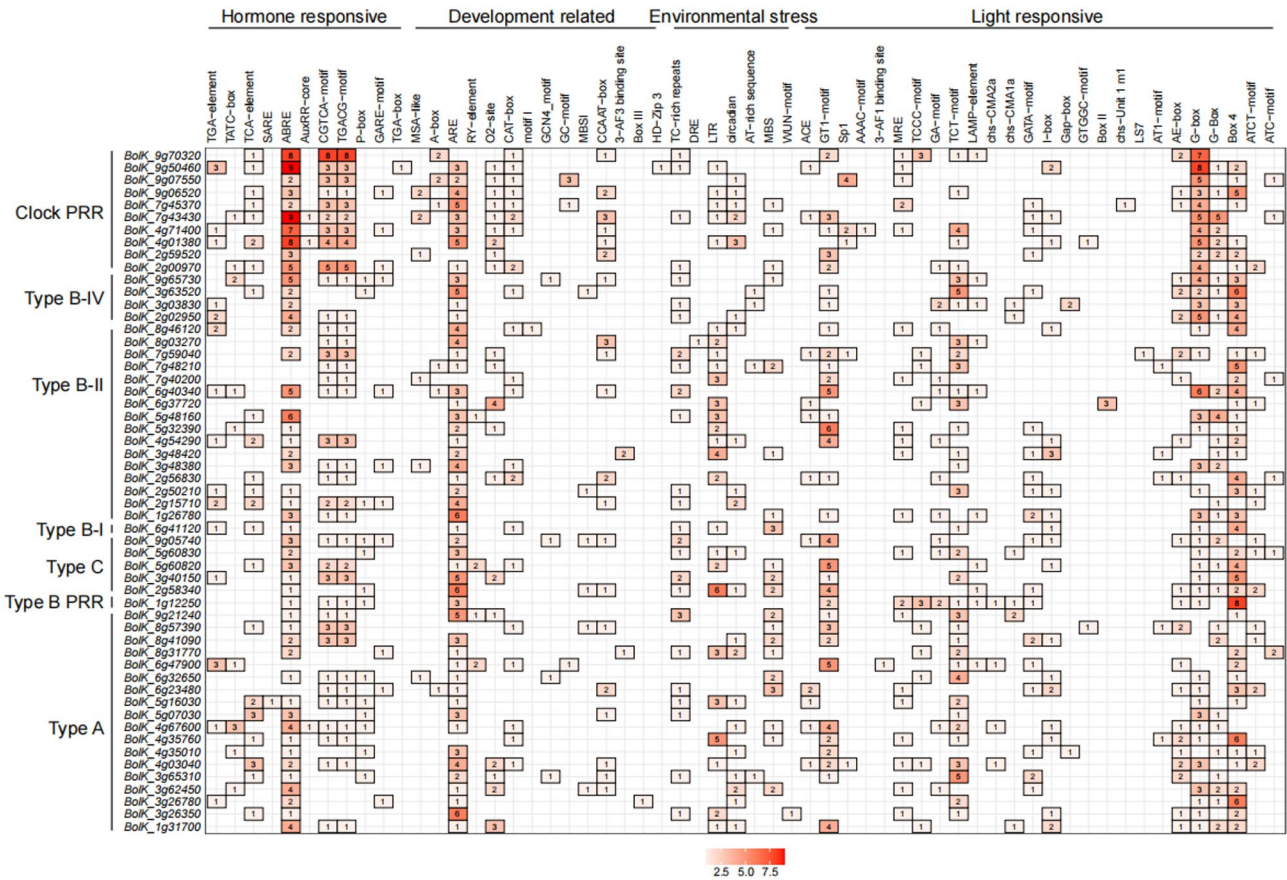


Fig. 4 Analysis of promoter sequence elements of the *BoRR* gene family in cauliflower. The ordinate represents each *BoRR* gene, and the abscissa represents the names of promoter elements related to hormones responsive, development related, environmental stress and light responsive. The numbers in the figure indicate the number of corresponding promoter elements contained in the promoter of the gene

and ethylene (Fig. 5A, Additional file: Table S3). Furthermore, this gene family is capable of responding to environmental signals like red light responses. It also plays a crucial role in influencing the growth and development of cauliflower. For example, it is involved in seed growth, meristem maintenance, and plant organ senescence pathways. In addition, it can affect the synthesis and metabolism of pigments. It is important to note that there are certain connections among these signal pathways, with some signal genes being shared among them (Fig. 5B). These shared genes may act as key regulatory elements, coordinating the complex interactions between different physiological processes.

Expression profiling of the *BoRR* gene family during cauliflower growth, curd development, and salt stress treatment

Firstly, the tissue specific expression of the *BoRR* gene family members in cauliflower was analyzed (Fig. 6A, Additional file: Table S4). The analysis results demonstrated that each type of the *BoRR* gene family in cauliflower exhibits distinct expression specificity. Type A *BoRR* genes (e.g., *BoRR13/19/23*) display a tendency to

have relatively high levels of expression in the roots and stems. Type C (e.g., *BoRR9/14/31/32*) and type B-IV (e.g., *BoRR5/11/18/56*) *BoRR* genes show a different pattern. They have higher expression levels in siliques. The clock PRR type *BoRR* genes (e.g., *BoRR41/52/53*) exhibit higher expression in flowers, buds, and siliques.

The expression characteristics of the *BoRR* gene family members during the formation process of the cauliflower curd were conducted. The analysis results indicated that some of the *BoRR* genes in cauliflower could respond to the curd differentiation process (Fig. 6B, Additional file: Table S4). During the transition stage from vegetative growth to reproductive growth, some type A *BoRR* genes (e.g., *BoRR23/27/34/38/45*) show high level expression. As the curd formation stage commences, some type A *BoRR* genes still maintain relatively high level expression. When it comes to the curd enlargement stage, several *BoRR* genes, such as *BoRR3*, *BoRR6*, *BoRR12*, *BoRR32*, and *BoRR54*, exhibit high level expression. Subsequently, during the transformation stage of the curd into flower buds, *BoRR* genes like *BoRR49* show high level expression.

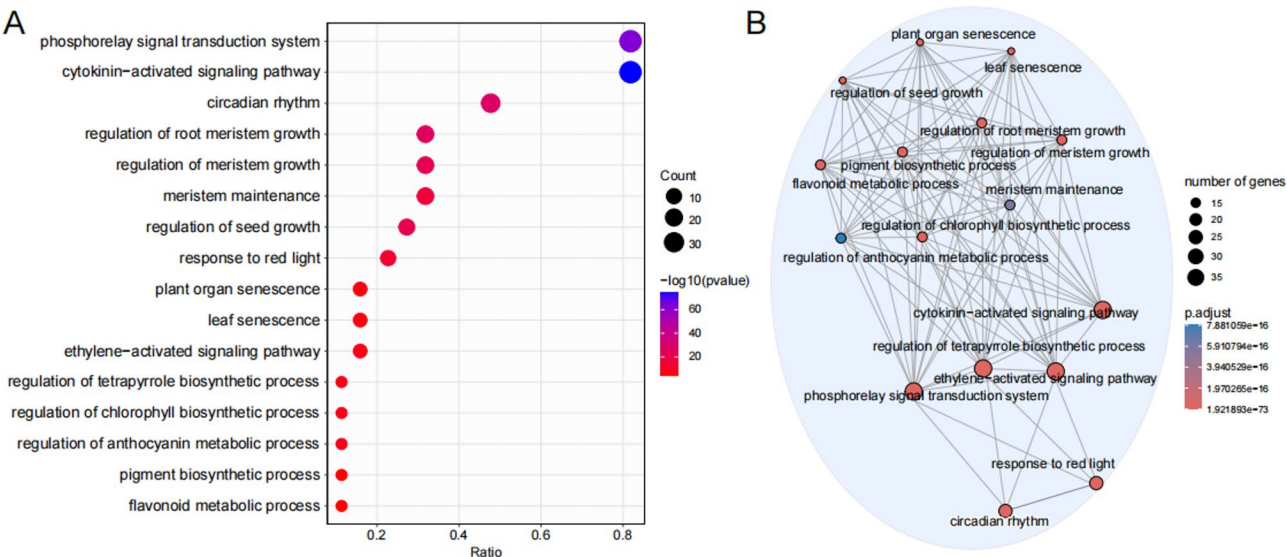


Fig. 5 Functional annotation of the *BoRR* gene family in cauliflower. **(A)** GO enrichment analysis of the *BoRR* genes in cauliflower. The abscissa represents the ratio of the enriched genes to the genes in the corresponding pathway, and the ordinate represents the names of the GO terms. The size of the dots indicates the number of genes, and the color of the dots represents the $p\text{-value}$. **(B)** Analysis of the correlation of the GO enrichment pathways of the *BoRR* genes in cauliflower. The size of the dots indicates the number of genes, and the color of the dots represents the adjusted $p\text{-value}$. The lines indicate the correlations

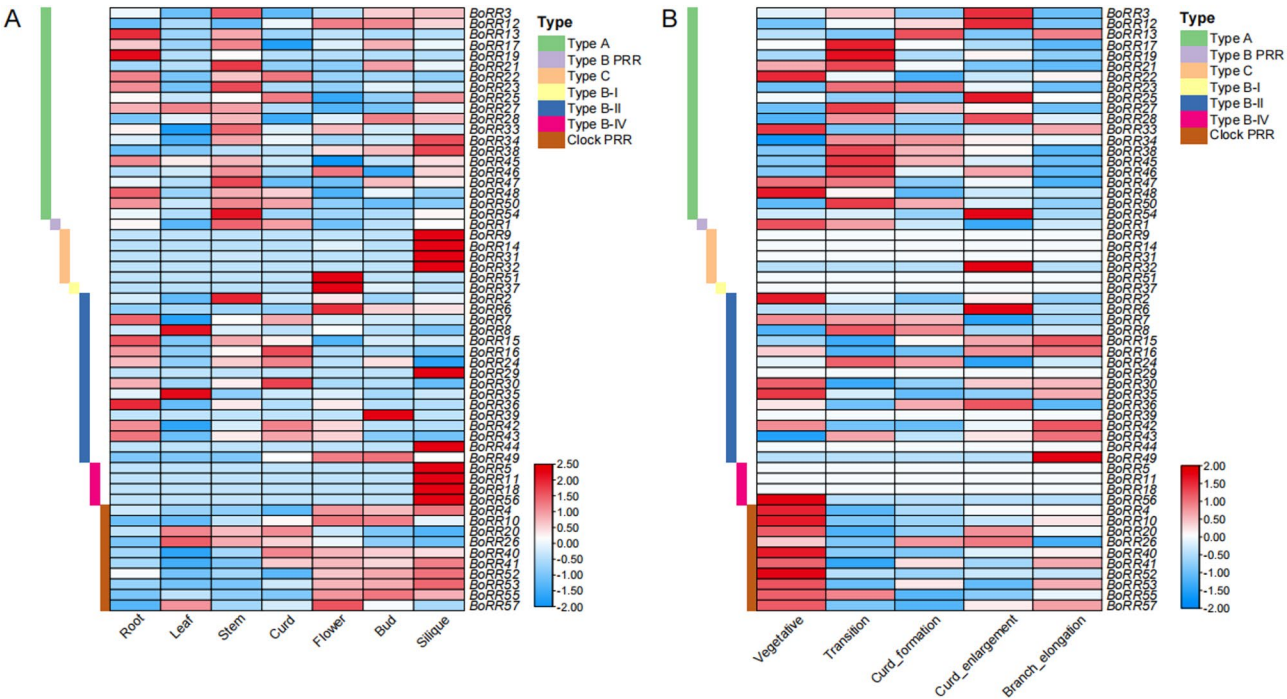


Fig. 6 Expression analysis of the *BoRR* gene family in cauliflower. **(A)** Heatmap of the expression of the *BoRR* gene family in different tissues of cauliflower. **(B)** Heatmap of the expression of the *BoRR* gene family at various stages of cauliflower curd development. Different colors marked on the left indicate different *BoRR* types. The expression values were standardized using the FPKM normalization method. Moreover, the z-score normalization was further applied along the horizontal axis. Red represents high expression, white represents medium expression, and blue represents low expression

We also used qRT-PCR to detect the expression profiles of some *BoRR* genes during various stages of cauliflower curd development (Fig. 7). Several genes, such as *BoRR1*, *BoRR4*, and *BoRR43*, display high level expression

during the stage when the shoot apical meristem (SAM) transforms into a curd. *BoRR45* shows high level expression in the initial curd formation stage. *BoRR23*, *BoRR30*, and *BoRR55* exhibit high level expression during the curd

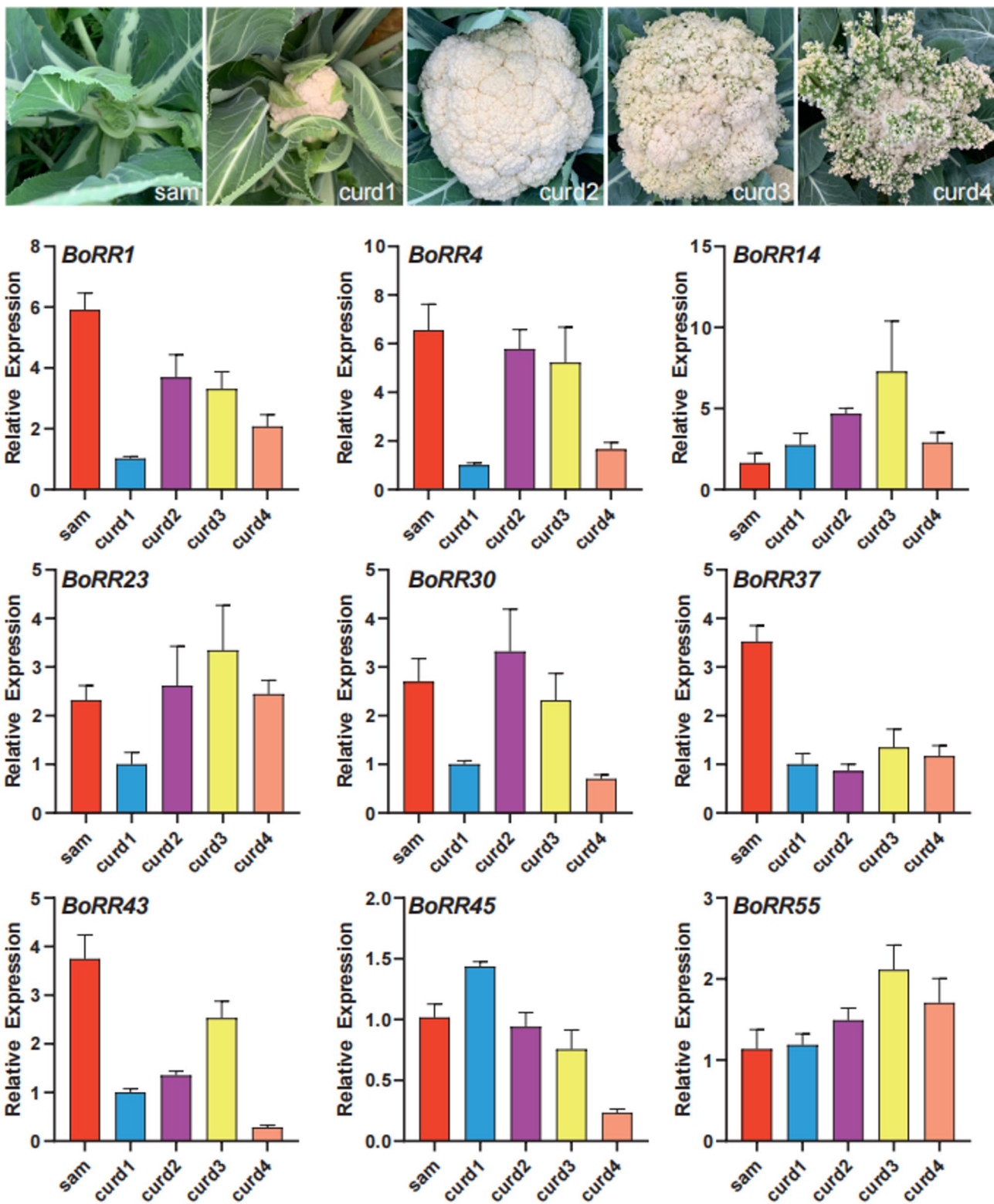


Fig. 7 Validation of the expression of *BoRR* genes during the development of cauliflower curds by qRT-PCR. The upper figure illustrates five stages of cauliflower curds: the apical meristem development stage, curd initiation, curd development, curd enlargement, and branch elongation. The lower figure shows the expression profiles of 9 *BoRR* genes across these 5 stages. The expression values were calculated by the $2^{-\Delta\Delta Ct}$ method

enlargement process. Similarly, *BoRR23* and *BoRR55* also have high level expression in the inflorescence growth stage.

Our study also investigated the functional involvement of the *BoRR* gene family in salt tolerance mechanisms of cauliflower through NaCl treatment experiments at seedling stage. As shown in Fig. 8A, salt-treatment cauliflower plants exhibited mild reduction in plant Height and slight leaf curling after 7 days of treatment while maintaining healthy growth compared to controls. Expression profiling of *BoRR* genes during salt stress (0, 1, 3, and 7 days post-treatment) revealed distinct temporal patterns. 8 of 9 selected *BoRR* genes displayed transient up-regulation with sharp expression increases at day 1, followed by gradual declines To baseline levels at days 3 and 7. Notably, *BoRR23* displayed a unique expression profile characterized by minimal fluctuation during early stress phases and significant down-regulation only at the 7-day time point.

Discussion

This study comprehensively analyzed the cauliflower *BoRR* gene family from multiple dimensions, including gene identification, classification, molecular characteristics, structural features, promoter elements, collinearity, GO enrichment and functional correlation, as well as the expression profiles during cauliflower growth, curd development and salt stress. Similar to the research findings on *BoRR* genes in *Arabidopsis* [23, 24], different types of *BoRR* genes in cauliflower may play diverse functional roles in its growth and development. For example, type A *BoRR* genes resemble *Arabidopsis* *ARR5/7* and rice *OsRR1* in conserved domains. *ARR5/7* negatively regulate cytokinin signaling via phosphorylation-dependent mechanisms, while rice *CRL5* induces *OsRR1* to repress cytokinin signaling during crown root initiation [13, 25]. These parallels suggested *BoRR* genes may share conserved roles in cytokinin-mediated cellular processes through analogous type-A RR regulation. Type B RRs in *Arabidopsis thaliana* (e.g., *ARR1/10/12*), rice (e.g., *OsRR21/22/23*), and *Brassica napus* (e.g., *BnARR12/14/18/23/33*), mediate complex signal cascades in cytokinin transduction, regulating growth processes like shoot/root development and inflorescence architecture, highlighting its conserved roles in hormone crosstalk and stress adaptation across species [6, 12, 26]. Type Clock PRR genes are conserved across plants and directly regulate circadian rhythms, as seen in *Arabidopsis* *TOC1/PRR1* forming feedback loops and *PRR5/7* repressing clock genes [20, 21]. Rice orthologs like *OsPRR9/5* integrate circadian signals for photoperiodic flowering [22], while their conserved pseudo-receiver and CCT domains enable nuclear protein complexes critical for oscillator function [11, 21].

It was predicted that most cauliflower *BoRR* proteins are concentrated in the nucleus, which is consistent with the localization of many transcription factors and signal transduction related genes (Table 1). For example, research on similar nuclear localized RR protein in rice has found that they play a crucial role in regulating the transcription process of genes related to rice growth and development [12, 22, 26]. The unique localization such as *BoRR5* on the plasma membrane, *BoRR14* in the chloroplast, and *BoRR8* in the endomembrane system reflect the specificity of the *BoRR* gene family's functions.

There are significant differences in motifs and domains among different classifications of *BoRR* gene families. The REC type domain contained in all cauliflower *BoRR* proteins is an important basis for their functional exertion (Fig. 2). The unique domain characteristics of different types of *BoRR* gene families determine their functional specificity. For example, the REC_typeA_ARR domain of type A may have a unique pattern in binding to specific upstream signal molecules and regulating downstream genes. Similar to the specific role of genes with similar domains in the cytokinin signal transduction pathway in *Arabidopsis* [13, 15]. Type B ARR (e.g., *ARR1/2*) with REC_typeB_ARR domains mediate cytokinin signaling via phosphorelay, while their GARP domains enable nuclear localization and transcriptional activation in meristems and vascular tissues [27]. Clock PRRs (e.g., *PRR5/9*) possess psREC_RR domains lacking phosphorylation sites, acting as transcriptional repressors through LDLSLRR motifs and CCT domains to bind CCA1/LHY promoters and regulate circadian rhythms [28]. The exon number variation in the *BoRR* gene family may generate transcript diversity through alternative splicing (e.g., exon skip or intron retention), potentially leading to functional diversification via proteins with distinct domains/localization, while the short CDS feature of type A members suggests they encode small regulatory proteins/peptides involved in signal transduction or transcriptional regulation (Fig. 2).

The diverse distribution pattern of promoter elements of the cauliflower *BoRR* gene family reflects its wide range of functional regulatory scope (Fig. 4). In terms of hormone responsiveness, the hormone response elements enriched in the promoters of type Clock PRR *BoRR* genes are closely related to the crucial regulatory role of hormones in plant growth and development. Conserved cis-element variants in hormone response promoters regulate response magnitude and spatial patterns [29]. Like ABRE motif act as key switches in ABA-dependent/independent gene expression for stress and developmental responses [30]. In terms of development related aspects, the ARE elements carried by most *BoRR* cauliflower genes promoter indicated their extensive participation in the entire growth and development process.

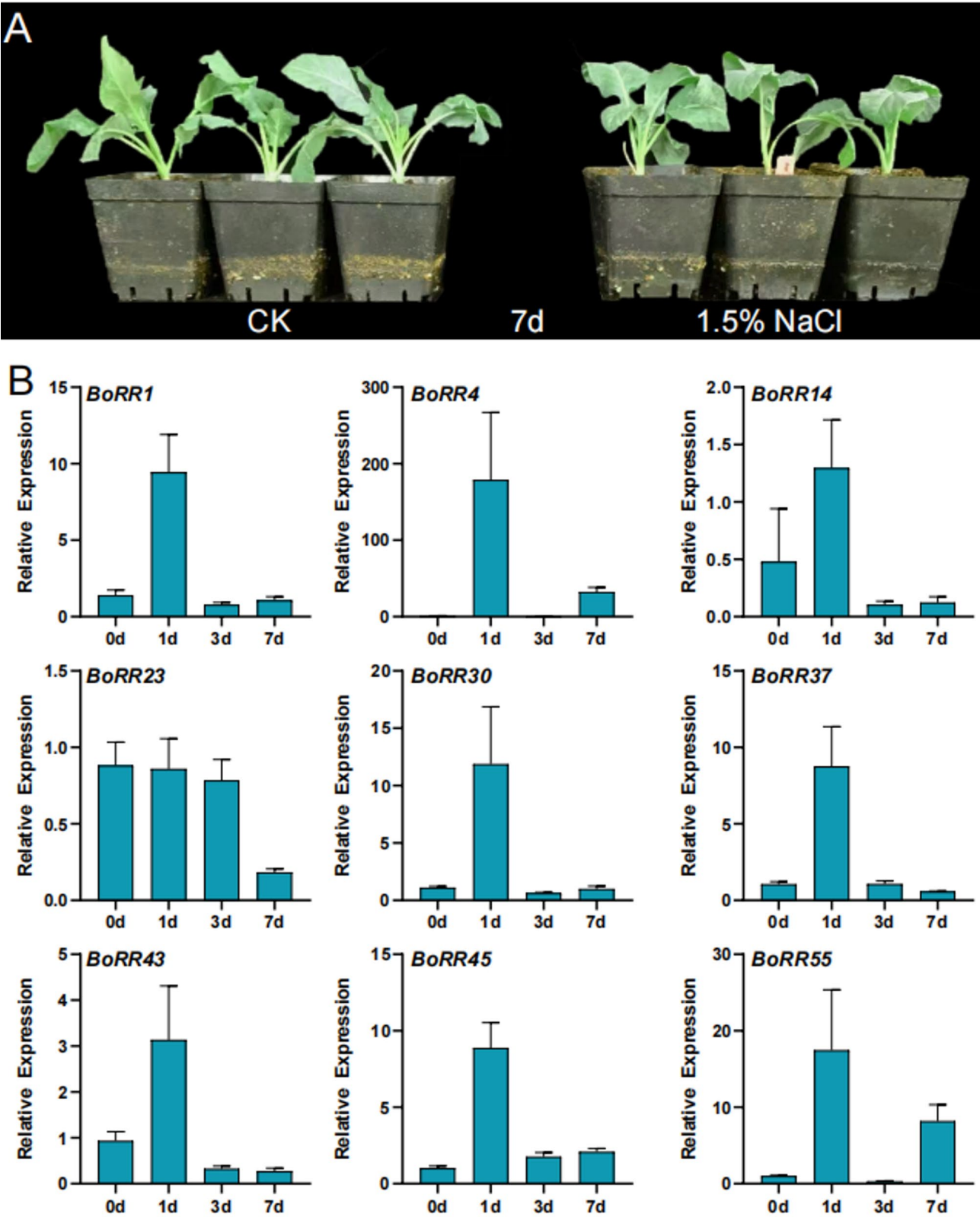


Fig. 8 Analysis of the expression levels of the RR gene family in cauliflower following salt stress treatment. **(A)** Phenotypes of cauliflower under normal conditions and after being treated with 1.5% NaCl for 7 days. **(B)** The expression levels of 9 *BoRR* genes in cauliflower at 0, 1, 3, and 7 days post-salt treatment. The expression values were calculated by the $2^{-\Delta\Delta Ct}$ method

From seed germination to organ senescence, *BoRR* genes may cooperate with many development related genes to jointly regulate the plant development process. In terms of environmental stress, the drought responsive elements of type B PRR and other types of genes reflect their importance in resisting drought environmental stress. These genes may play a key regulatory role, similar to the functions of drought responsive genes in wheat and other crops [17, 18, 20, 31]. In terms of light responsiveness, the light response elements of some type A *BoRR* genes have an impact on the photomorphogenesis of cauliflower. Type A *BoRR* genes may regulate the physiological processes related to the photomorphogenesis of cauliflower by responding to light signals, such as the regulation of stem elongation inhibition, leaf expansion and chloroplast development [19, 32, 33].

The collinearity analysis results of cauliflower *BoRR* genes show complex events in its genome evolution (Fig. 3A). The numerous collinear relationships across chromosomes indicated that frequent inter chromosomal interactions and gene recombination events have occurred during the genome evolution of the *Brassica* genus. The small number of collinear gene pairs within chromosomes reflected that the gene structure within chromosomes is relatively stable and may be related to maintaining the basic functions of chromosomes. The one to many collinear relationships of some genes are closely related to the triploidization of the genome of the *Brassica* genus [34–36].

The collinearity analysis results of cauliflower with other species revealed *BoRR* gene family phylogenetic relationship and gene conservation and divergence during evolution (Fig. 3B). There are more shared collinear gene pairs with *Brassica napus* and *Brassica rapa*, since they all belong to the *Brassica* genus. There are also a certain number of collinear gene pairs with other dicotyledonous plants such as *Arabidopsis* and *Glycine max*, reflecting a certain phylogenetic relationship and partial conservation of gene functions among dicotyledonous plants during evolution. Compared with *Cucumis melo*, which has a relatively distant evolutionary relationship, and monocotyledonous plants such as *Oryza sativa* and *Zea mays*, the number of collinear gene pairs is relatively small and only one pair of genes is collinear with monocotyledonous plants, indicating that the *BoRR* genes in cauliflower have a closer relationship with dicotyledonous plants in the same or related genera and have significant divergence from the *BoRR* genes in monocotyledonous plants during evolution.

The expression specificity of different types of members of the cauliflower *BoRR* gene family in different tissues reflected their functional division (Fig. 6A). The high expression of type A *BoRR* genes in roots and stems may be related to the physiological function requirements of

roots and stems. The high expression of type C and type B-IV *BoRR* genes in siliques implies their crucial roles in silique development, seed formation and maturation. Type C and type B-IV *BoRR* genes may cooperate with genes related to seed nutrient accumulation, silique cell wall metabolism and seed dormancy regulation. The high expression of type Clock PRR *BoRR* genes in flowers, buds and siliques indicates their multifaceted roles in flower development, flowering time regulation and the later development and maturation of siliques.

The expression patterns of *BoRR* genes during cauliflower curd formation highlighted their regulatory roles at various stages (Figs. 6B and 7). In the transition from vegetative to reproductive growth, high expression of type A *BoRR* genes likely regulates hormone balance, signal transduction, and gene expression reprogramming. During curd initiation, these genes may promote curd primordia formation. In the curd enlargement stage, genes like *BoRR3*, *BoRR6*, *BoRR12*, *BoRR32*, and *BoRR54* likely regulate cell division, elongation, and nutrient transport, supporting curd growth. In the transition from curd to flower buds, *BoRR49* and other genes may assist in flower bud formation by influencing differentiation and hormone signaling. These coordinated gene actions contribute to the intricate processes of curd development and flower bud transformation. Understanding these functions could enhance cauliflower cultivation and yield.

As central regulators of plant stress responses, the *RR* gene family dynamically coordinates plant growth and stress adaptation by integrating environmental signals with endogenous hormonal pathways [12, 13]. Certain plant gene families exhibit responses to salt stress. For instance, the *GATA* gene family plays a part in enhancing the salt tolerance of onions [37], the *LRX* gene family in Cucurbitaceae can respond to salt stress and drought stress in cucumber [38], the *AgSK22* gene belonging to the *GSK3* gene family in celery is capable of responding to salt stress [39], and the *FAH* gene family contributes to improving the salt tolerance of cotton [40]. Previous studies have demonstrated the inherent salt tolerance of *Brassica oleracea* vegetables, yet their molecular regulatory networks remain incompletely characterized [41, 42]. Our research revealed that the rapid up-regulation of *BoRR* family members during the early salt stress phase (1 day post-treatment) likely contributes to initial stress defense through activation of signal transduction pathways or osmoprotective mechanisms, while the progressive down-regulation of most genes during mid-to-late stress stages (3–7 days) suggested phase-dependent functional shifts during salt adaptation (Fig. 8). The dynamic expression patterns suggested that the *BoRR* family may regulate the shift from stress responses to homeostasis by coordinating processes like ion balance and metabolic

changes, providing new insights into the polygenic mechanisms of salt tolerance.

The study's reliance on in silico predictions and expression data may limit the accuracy of functional insights of *BoRR* genes. Further experimental validation is needed to confirm the roles of these genes in developmental regulation, curd formation and stress tolerance of cauliflower.

Conclusion

This study analyzed the Cauliflower *BoRR* gene family. A Total of 57 *BoRR* genes were identified and classified. Their molecular characteristics are diverse, and their sub-cellular localizations are different, reflecting functional differences. Structural analysis uncovered the characteristics of gene family members in motifs, domains, and gene structures. The domains of different types of genes are related to specific functions. Promoter element analysis indicated that *BoRR* genes respond differently to hormones, development, environmental stress, and light, demonstrating their roles in regulatory mechanisms. Collinearity analysis showed that complex inter-chromosomal interactions occurred during the genome evolution of cauliflower. There are more collinear genes with other *Brassica* species, and significant differences were observed compared with monocotyledonous plants. GO enrichment analysis revealed that this gene family is involved in multiple signal transduction pathways and growth and development processes, and there are key genes coordinating different physiological processes. Expression profiling analysis suggested that different types of *BoRR* genes have specific expression patterns in various tissues and different stages of cauliflower curd development, playing diverse roles in curd initiation, growth, and the transition to inflorescence. Notably, several *BoRR* members showed salt stress-inducible expression profiles, aligning with their potential involvement in abiotic stress adaptation. The present study has provided insights into the properties and potential functional mechanisms of the *BoRR* gene family in cauliflower.

Materials and methods

Plant material

The cauliflower inbred line “3231” was developed by the Institute of Vegetables of Zhejiang Academy of Agricultural Sciences, with previous experiments confirming its strong salt Tolerance with served as the basis for its selection in this study. The experiment was carried out at the Zhejiang Academy of Agricultural Sciences, where seedlings were cultivated under controlled conditions of 22 °C and a 12-hour light/dark cycle until the three-leaf stage. At this stage, the seedlings were transplanted and divided into two groups. The control group was treated with standard Hoagland nutrient solution, while the salt-treated group was irrigated with a solution supplemented with

1.5% NaCl (w/v) every 48 h. The 1.5% NaCl concentration was determined through preliminary experiments, which showed that it could effectively induce salt stress and was suitable for evaluating the salt tolerance of “3231”. The Hoagland nutrient solution was prepared according to the formula described in Hoagland and Arnon (1950) [43]. The salt treatment started from the three-leaf stage and lasted for 7 days, with samples collected at 0, 1, 3, and 7 days. Three biological replicates were set up, with 6 plants in each replicate.

Genome-wide identification of *BoRR* gene family in cauliflower

First, the “Korso” cauliflower reference genome was downloaded from BoGDB [44], including genome sequences, coding sequences, protein sequences, and annotation files. Concurrently, the RR protein-specific HMM seed file (the model file containing conserved motifs/statistical parameters for RR proteins, PF00072 from Pfam) was obtained from Pfam. Using HMMER3.1's *hmmsearch* with default parameters, the protein database was screened with this seed file [Data: March, 2024]. The initial *BoRR* protein sequences were then subjected to multiple sequence alignment via ClustalW, followed by building a cauliflower-specific *BoRR* HMM model using *hmmbuild* [Data: March, 2024]. A second *hmmsearch* was performed with this model, and proteins lacking the core *BoRR* domain (validated via NCBI CDD) were manually excluded to obtain the final *BoRR* gene family [Data: April, 2024].

Chromosomal location, protein characteristics and structure analysis of *BoRR* gene family in cauliflower

The chromosomal location information of the *BoRR* gene family in cauliflower was procured with the gene annotation of the cauliflower genome database and graphically represented using Circos software (<http://circos.ca/>) [Data: April, 2024]. By searching the cauliflower genome file according to the gene name, the full-length protein sequences of the *BoRR* members were obtained. The online sequence analysis tool software (http://www.detaibio.com/sms2/protein_mw.html) was then utilized to compute the protein molecular weight and isoelectric point information (PI value) [Data: April, 2024]. The online software WOLF PRORT (<https://www.genscript.com/wolf-psort.html?src=leftbar>) was employed to predict the protein subcellular localization [Data: May, 2024]. The MEME online software (<http://meme-suite.org/tools/meme>) was engaged To analyze the protein Motif characteristics, with the number of motif searches set to 10 and other parameters left at default values [Data: May, 2024].

Evolutionary analysis of BoRR gene family in cauliflower and *Arabidopsis*

The protein sequences of the *Arabidopsis* ARR gene family were downloaded from the *Arabidopsis* database (<https://www.arabidopsis.org/>). A clustering analysis of the cauliflower and *Arabidopsis* ARR gene families was carried out using the Neighbor joining method, and a phylogenetic tree was constructed using MEGA7 software [Data: May, 2024]. The Bootstrap value was configured as 1000, and other parameters are set to default. The graphical display was adjusted using the online software EvolView (<https://www.evolgenius.info/evolview/>) [Data: May, 2024]. Based on homology with *Arabidopsis* proteins, the BoRR genes were classified into different types [Data: May, 2024].

Cis-regulatory elements analysis

Promoter regions (2000 bp upstream of transcription start sites) of BoRR genes in cauliflower were extracted from genomic data [Data: February, 2025]. Using PlantCARE, elements related to hormone response, development related, environmental stress, and light responsiveness were identified and counted. The distribution and quantity of these elements were visualized to analyze their functional implications [Data: February, 2025].

Collinearity analysis of RR gene family

For intra-genomic collinearity, cauliflower BoRR protein sequences were blasted against the genome using MCS-canX (E-value < 1e-10), with parameters set for match score (50) and gap penalties. Collinear pairs were visualized via Circos, distinguishing inter-chromosomal and intra-chromosomal relationships [Data: June, 2024]. For interspecific analysis, the same MCS-canX pipeline was applied to compare BoRR genes with *Arabidopsis*, *B. rapa*, *B. napus*, soybean, melon, rice, and maize, counting collinear pairs to assess evolutionary relationships [Data: July, 2024].

GO enrichment and functional correlation analysis

BoRR protein sequences were annotated via InterPro to derive GO terms. ClusterProfiler in R was used for enrichment analysis, identifying significant terms in signal transduction, environmental response, and developmental processes. Pathway correlations were inferred from shared genes, visualized as a network to show regulatory intersections [Data: September, 2024].

Expression analysis of BoRR gene family in cauliflower

The raw transcriptome data of cauliflower protein-coding genes in various tissue expression analyses were downloaded from the NCBI SRA database (<https://www.ncbi.nlm.nih.gov>, Accession numbers: PRJNA546441). Salmon

software (version 1.10.2, <https://github.com/COMBINE-lab/salmon>) was used to calculate the expression levels of all genes in cauliflower, with alignment performed against the cauliflower “Korso” genome. The expression levels were normalized using FPKM values. Subsequently, the expression levels of genes from the BoRR gene family in various tissues of cauliflower were extracted, and a heat map of these expression levels was generated using TBtools software for visualization [Data: from October To November, 2024].

Total RNA was extracted from different curd development stages and salt treatment stages (0d, 1d, 3d, and 7d) using the RNA extraction kit (TaKaRa, Code No. 9767; <https://www.takarabiomed.com.cn/>). Reverse transcription and cDNA synthesis were performed using the PrimeScript™ 1st Strand cDNA Synthesis Kit (TaKaRa, Code No. 6110 A; <https://www.takarabiomed.com.cn/>). Quantitative real-time PCR was performed using the TB Green Premix Ex Taq™ kit (TaKaRa, Code No. RR820A; <https://www.takarabiomed.com.cn/>) following the manufacturer's instructions on an Applied Biosystems StepOnePlus™ Real-Time PCR System. Melting curve analysis was conducted to verify primer specificity, ensuring single product amplification without nonspecific binding. A Total of 9 genes, representing different subgroups of the cauliflower BoRR gene family, were selected for qPCR validation. The actin gene was used as the reference gene for normalization. Three biological replicates were used for each treatment, with technical replicates included for each biological sample. Gene expression was calculated using the $2^{-\Delta\Delta Ct}$ method. The relevant primers are listed in Table S5.

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12864-025-12005-x>.

Supplementary Material 1.

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Not applicable.

Authors' contributions

M.S. designed and conducted the experiments. M.S. and Y.S. conducted bioinformatics analysis and wrote the manuscript. J.W. and H.Y. contributed plant materials, performed the experiments and analyzed the data. X.S. and S.Q. sorted out the data and revised the article. H.G. the corresponding author, conceived and designed the experiments, and revised the manuscript. All authors have approved the manuscript.

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Data availability

All data generated or analysed during this study are included in this published article and its supplementary information files. The transcriptome data analyzed in this article are from the public database NCBI, with the accession number: PRJNA546441.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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